

From Regulatory Text to Compliance Knowledge Graphs: Extraction, Conflict Detection, and Normative Gap Analysis Across Trustworthy AI Standards

Anonymous submission

Abstract

Trustworthy AI regulations — including the EU AI Act, ISO/IEC 42001, and ISO/IEC 5259-3 — impose legal and normative obligations on AI systems in natural language that is too broad and socio-technical to operationalize directly. Existing text-to-graph extraction approaches extract facts in isolation, discard deontic obligation strength, and produce graphs with no memory of what has already been captured. We present **KBDebugger**, a human-in-the-loop pipeline that transforms regulatory documents into structured, provenance-tracked knowledge graphs through seven sequential stages: keyword-guided subgraph retrieval, structure-aware document parsing, atomic quality decomposition, node-entity similarity filtering, LLM-based novelty classification, controlled-vocabulary triplet extraction, and expert review. The pipeline tracks deontic modality (MANDATORY, RECOMMENDED, OPTIONAL, PROHIBITED) for each extracted triple and supports multi-document analysis for obligation conflict detection, normative gap identification, and cross-standard coverage analysis. On the MINE-1 knowledge retention benchmark (100 articles, 15 facts each), KBDebugger achieves **79.4%** retention versus 62.7% for KGen — a +16.7 point improvement with the same retriever and judge. Applied to three co-regulating Trustworthy AI standards, the system surfaces obligation conflicts and undefined-concept gaps that document-level reading cannot scale to detect.

1 Introduction

The legal operationalization of AI governance has become one of the most pressing engineering challenges in applied AI. Regulations such as the EU AI Act (European Parliament and Council of the European Union 2024), ISO/IEC 42001 (International Organization for Standardization 2023), and ISO/IEC 5259-3 (International Organization for Standardization 2024) each define obligations for AI systems across dimensions including transparency, fairness, robustness, privacy, and human oversight. Yet these standards are written in normative natural language — using hedged modal constructions (*shall*, *should*, *may*) and referencing concepts they seldom formally define — making it exceptionally diffi-

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cult for engineers and compliance teams to derive concrete, actionable requirements.

The challenge is compounded by scale: a compliance review team working across three co-regulating standards faces thousands of normative assertions, many of which nominally address the same concept (e.g., *data quality*, *transparency mechanism*) but with differing obligation strengths, subtly conflicting scopes, or mutually inconsistent definitions. Manual cross-reading cannot sustain this at scale, and the growing body of Trustworthy AI regulation makes the problem structurally worse over time.

Existing NLP and knowledge engineering approaches fall short in three ways. First, text-to-graph extractors such as KGen (Mo et al. 2025), GraphRAG (Edge et al. 2024), and OpenIE-based systems extract triples from individual documents without conditioning extraction on what is already known — every run re-derives the same facts and lacks a deduplication model. Second, regulatory NLP tools (Abualhaija et al. 2025; Guliani et al. 2026) focus on obligation detection or deontic classification within a single document but do not integrate extracted knowledge into a growing, queryable graph. Third, none of these approaches tracks deontic modality at the triple level, making it impossible to detect a central class of inter-standard conflict: one standard mandating what another merely recommends for the same claim.

We address these gaps with **KBDebugger**, a seven-stage human-in-the-loop pipeline that treats incremental knowledge graph construction as a first-class design concern. The pipeline conditions each extraction step on the current graph state, classifies new information by novelty relative to existing knowledge, records deontic obligation strength per triple, and exposes cross-document conflicts and coverage gaps through a structured comparison engine. The system is evaluated both quantitatively on the MINE-1 knowledge retention benchmark and qualitatively through multi-standard analysis of three Trustworthy AI regulatory documents.

This paper is organized around five research questions:

RQ1 *Normative gap exposure*: When Trustworthy AI standards impose obligations on concepts they never formally define, what is the scale and distribution of these normative gaps, and can structured KG extraction surface them more systematically than document-level reading?

- RQ2** *Cross-standard concept coverage:* What fraction of the Trustworthy AI concept space is jointly covered by multiple standards versus exclusively asserted by one, and do coverage asymmetries emerge across dimensions such as fairness, transparency, and data governance?
- RQ3** *Human oversight contribution:* What classes of LLM extraction error does expert review correct in a regulatory triplet pipeline, and does the correction pattern reveal structurally different failure modes in deontic (normative) versus descriptive (definitional) regulatory text?
- RQ4** *Obligation conflict detection:* Across co-regulating Trustworthy AI standards, to what extent do structurally identical claims carry conflicting deontic obligations (MANDATORY vs. RECOMMENDED vs. OPTIONAL), and can provenance-tracked, modality-tagged KG extraction surface these tensions automatically?
- RQ5** *Knowledge retention:* Does a novelty-aware, ontology-constrained extraction pipeline achieve higher knowledge retention on the MINE-1 benchmark than automated methods (KGGen, GraphRAG, OpenIE), and which pipeline stages contribute most to the improvement?

This paper makes the following contributions:

1. A seven-stage human-in-the-loop pipeline for incremental, graph-conditioned extraction of regulatory knowledge from Trustworthy AI standards, with deontic modality tracking per triple and append-only cross-document provenance.
2. A novelty-aware classification mechanism (EXISTING / PARTIALLY_NEW / NEW) that conditions LLM extraction on the current knowledge graph state, reducing redundancy and improving the semantic density of extracted knowledge.
3. A multi-document comparison engine that detects four classes of inter-standard conflict (obligation, definition, taxonomy, value), surfaces normative gaps (obligations on undefined concepts), and produces per-dimension coverage matrices across standards.
4. A pre-merge KB preview and five-strategy post-upsert verifier (Coverage, Correctness, Consistency, Completeness, Minimality) that close the quality loop — detecting extraction errors that survive human review before the graph is used for compliance reasoning.
5. An empirical evaluation on the MINE-1 benchmark demonstrating a +16.7 point knowledge retention improvement over KGGen, and a qualitative multi-standard analysis of the EU AI Act, ISO/IEC 42001, and ISO/IEC 5259-3.

2 Background and Related Work

2.1 Text-to-Knowledge Graph Extraction

Automated knowledge graph construction from text has evolved through three generations: rule-based OpenIE (Banko et al. 2007; Niklaus et al. 2018), neural entity-relation extraction (Huguet Cabot and Navigli 2021; Ye et al.

2022; Tang et al. 2022), and, most recently, LLM-driven graph generation (Zhu et al. 2024; Pan et al. 2024; Zhang and Soh 2024; Xu et al. 2024).

Mo et al. (2025) introduce KGGen alongside the MINE benchmark — the first quantitative benchmark for text-to-KG knowledge retention — and serve as our primary comparison point. Edge et al. (2024) build community-level graphs for retrieval-augmented generation; their approach emphasizes global summarization rather than provenance fidelity. Wadhwa, Amir, and Wallace (2023) demonstrate that GPT-class models already outperform fine-tuned specialist models on standard RE benchmarks, motivating KBDebugger’s LLM-based extraction stages. Zhang and Soh (2024) propose the Extract-Define-Canonicalize framework, which parallels KBDebugger’s controlled-vocabulary design and canonicalization step. A comprehensive survey of LLM-based generative IE (Xu et al. 2024) and a survey of neural OpenIE (Zhong et al. 2022) situate KBDebugger within the broader landscape.

2.2 Regulatory and Legal NLP

Legal NLP has progressed from domain-adapted language models such as LEGAL-BERT (Chalkidis et al. 2020) through structured compliance checkers and, most recently, to LLM-driven regulatory extraction. Sleimi et al. (2018) extract semantic legal metadata from EU legislation using dependency parsing and deontic keyword patterns, establishing a precedent for the structured normative extraction KBDebugger performs. Dragoni et al. (2019) combine constituency parsing with ML classifiers to extract rules (obligations, permissions, prohibitions) from legal text — the same three-class deontic scheme KBDebugger’s predicate vocabulary encodes as MANDATORY / RECOMMENDED / OPTIONAL / PROHIBITED.

Deontic modality detection. Detecting modal obligation strength (*shall*, *should*, *may*) in legal text is a distinct NLP task. Sakketou, Chalkidis, and Roussinov (2022) introduce a dataset annotated for agent-specific deontic modality in legislation; Liasetsanagakis and Chalkidis (2023) evaluate LEGAL-BERT for deontic classification in EU legislation under low-resource conditions. Both establish baseline performance for the modality classification KBDebugger applies in Stage 5.

Automated compliance checking. Amaral et al. (2023) achieve 89.1% precision extracting “shall” requirements from GDPR data processing agreements — the closest structural analogue to KBDebugger applied to a single standard. Abualhaija et al. (2025) extend this to LLM-assisted GDPR requirement extraction. Guliani et al. (2026) apply iterative LLM self-refinement for structured regulatory rule extraction. Feng et al. (2024) operationalize normative requirements with LLMs at IEEE RE 2024, directly paralleling KBDebugger’s Stage 5 triplet extraction with deontic labels. Colombo, Rognone, and Palmirani (2025) apply an NLP pipeline to obligations in the EU AI Act, DSA, and GDPR — the same regulatory corpus KBDebugger targets.

A recent survey of legal NLP (Ariai and Demartini 2024) and the LegalBench benchmark (Guha et al. 2023) (162 tasks measuring legal reasoning in LLMs) provide the evaluation

landscape within which KBDebugger’s extraction stages operate. ContractNLI (Koreeda and Manning 2021) demonstrates NLI as a tool for identifying obligations and entailments in legal contracts — directly relevant to KBDebugger’s novelty classification step (Stages 3–4).

2.3 Trustworthy AI Standards and Operationalization

The landscape of Trustworthy AI governance is fragmented. Jobin, Ienca, and Vayena (2019) survey 84 AI ethics guidelines, finding convergence on five principles (transparency, fairness, accountability, privacy, robustness) but divergence on operationalization — exactly the gap KBDebugger’s multi-standard comparison is designed to quantify. Mittelstadt (2019) argue that principles alone cannot guarantee ethical AI absent concrete technical mappings; Hagendorff (2020) reach the same conclusion from an empirical evaluation of 22 guidelines. Together they motivate KBDebugger’s normative gap detection: surfacing concepts standards oblige but never define.

The EU AI Act (European Parliament and Council of the European Union 2024) and ISO/IEC 42001 (International Organization for Standardization 2023) represent the most binding instruments at present. Laux, Wachter, and Mittelstadt (2022) analyse the EU AI Act’s risk-based trustworthiness framework, noting the unresolved tension between legal acceptability and technical operationalizability. Yang, Kim, and Yoon (2026) propose PASTA, a scalable multi-policy AI compliance evaluator using LLMs — the most comparable published system architecture to KBDebugger’s comparison engine. Ershov, Mitra, and Tiwari (2023) open-source a Neo4j-backed regulatory KG for Abu Dhabi financial regulations, demonstrating the same graph-plus-NLP pipeline design KBDebugger adopts for AI standards.

2.4 Human-in-the-Loop Knowledge Engineering

Human-in-the-loop (HITL) approaches consistently outperform fully automated KG pipelines on quality-sensitive tasks (Mosqueira-Rey et al. 2023). Bikaun, Stewart, and Liu (2024) present CleanGraph, an interactive CRUD tool for KG refinement and completion — the closest published system to KBDebugger’s review interface, though CleanGraph operates on the graph directly rather than at the pre-upsert triplet level. Tang et al. (2024) introduce CollabKG, a learnable human–machine cooperative IE tool with LLM pre-annotation and active self-renewal — the architecture most similar to KBDebugger’s Stage 6 reviewer workflow, where LLM-proposed triplets are presented to experts for correction before any graph write. Neither system exposes pre-merge conflict previews or post-upsert verification.

2.5 Knowledge Graph Quality, Provenance, and Verification

KG quality management encompasses accuracy, consistency, completeness, timeliness, and provenance (Xue and Zou 2023; Zaveri et al. 2016). Dibowski (2024) propose triple-level change tracking with full provenance for KGs, directly applicable to KBDebugger’s append-only provenance model.

The W3C PROV-O ontology (Lebo et al. 2013) provides the interoperability standard for provenance graphs that KBDebugger’s Neo4j data model aligns with. Meroño-Peñuela et al. (2025) argue that knowledge graphs should serve as the backbone of AI data governance — a proposition KBDebugger instantiates for regulatory compliance.

For KG evaluation, Chen et al. (2022) demonstrate that standard link prediction metrics systematically underestimate KG quality under the open-world assumption — motivating KBDebugger’s human-verified MINE evaluation rather than purely automated scoring. KBDebugger’s five-strategy verifier (S1 Coverage, S2 Correctness, S3 Consistency, S4 Completeness, S5 Minimality) extends the quality dimensions identified in these surveys to a runtime, read-only verification pass on the live graph.

2.6 Multi-Document Analysis and Conflict Detection

Detecting inconsistencies across documents or knowledge bases requires both semantic alignment of shared concepts and logical comparison of their asserted properties. Wang et al. (2024) propose knowledge-guided cross-document relation extraction at ACL 2024 — directly paralleling KBDebugger’s comparison engine, which must relate obligations from separate regulatory standards documents. Bassignana and Plank (2022) introduce CrossRE, a cross-domain RE benchmark exposing generalization gaps, motivating KBDebugger’s domain-specific predicate vocabulary for regulatory text.

For legal-domain conflict detection, Arora, Grundy, and Abdelrazek (2024) combine formal logic with LLMs to detect contradictions in requirements specifications — the same task KBDebugger’s conflict detection stage performs at the knowledge-graph level. Akasiadis et al. (2025) address automated KG inconsistency detection and repair, confirming that graph-level conflict reasoning is tractable and practical. Ontology alignment methods from the OAEI (Ontology Alignment Evaluation Initiative 2024) provide evaluation standards for the semantic alignment step in KBDebugger’s comparison engine, where SAME_AS pairs are proposed and reviewed before canonicalization.

2.7 Sentence Embeddings and Knowledge-Enhanced Representations

KBDebugger’s similarity filtering (Stage 3) relies on sentence embeddings from Sentence-BERT (Reimers and Gurevych 2019) (all-MiniLM-L6-v2), which produces dense representations suited to the short, label-style node names of ontology-style KGs. Devlin et al. (2019) provide the pre-training foundation for all domain-adapted models used in the pipeline; Zhang et al. (2019) demonstrate that integrating KG entity knowledge into LM pre-training improves factual recall — motivating KBDebugger’s graph-conditioned approach where the existing KG guides novel extraction rather than operating independently. Chain-of-thought prompting (Wei et al. 2022) is employed in KBDebugger’s novelty classification and conflict judgment prompts to encourage step-by-step reasoning over regulatory clause comparisons.

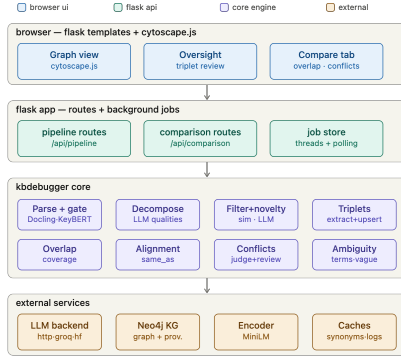


Figure 1: KBDebugger pipeline overview. Seven stages transform a regulatory PDF into provenance-tracked triples in a Neo4j knowledge graph, with human review at Stage 6 before any graph upsert.

3 System Design and Methodology

KBDebugger is a seven-stage pipeline for human-in-the-loop knowledge graph construction from regulatory documents. The system is grounded in three design principles: (i) *graph-conditioned extraction* — every filtering and classification step is informed by the current graph state, not applied in isolation; (ii) *deontic fidelity* — obligation strength is preserved as a first-class property of each triple, not normalized away; and (iii) *append-only provenance* — the full source context of every triple is retained regardless of how many subsequent documents re-assert the same claim.

Figure 1 provides an overview of the pipeline and its component stages.

3.1 Stage 1: Graph-Context Retrieval

Before any document is ingested, the system retrieves the relevant subgraph from the existing knowledge graph using keyword-guided pattern matching against node names, relation types, and stored provenance metadata in Neo4j. This subgraph serves as the semantic context against which new extractions are compared in Stages 3 and 4. The retrieval step ensures that extraction is not document-agnostic: the system knows what it already knows.

Retrieval is performed via three complementary patterns: keyword occurrence in source node name, in target node name, and in relationship type or sentence metadata. Results are deduplicated and limited to a configurable per-pattern count (default 50 relations per pattern) to prevent context bloat for densely interconnected topics.

3.2 Stage 2: Document Ingestion and Quality Decomposition

Stage 2 has three substages that transform a raw PDF into a set of *atomic quality sentences* — the granular unit of analysis throughout the remainder of the pipeline.

2a. Structure-Aware PDF Parsing. The system uses Docling (DS4SD Team 2024) for layout-aware PDF parsing with optional OCR and table recognition. A conservative

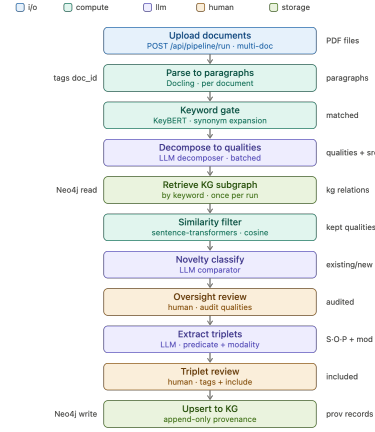


Figure 2: Stage 2 ingestion substages: PDF parsing (Docling), keyword-gate filtering (KeyBERT), and atomic quality decomposition (LLM). Each substage reduces the candidate set and increases semantic granularity.

post-processing pass filters reference sections to prevent bibliographic noise from entering the extraction stream. Each paragraph is retained with its full provenance (page number, heading hierarchy, document title).

2b. Keyword Filtering. Paragraphs are filtered by relevance to the target dimension using KeyBERT (Grootendorst 2020), a sentence-transformer-based keyphrase extractor. The system supports three filtering modes: (i) a curated Trustworthy AI dimension keyword (e.g., *fairness*, *robustness*); (ii) user-supplied custom keywords with LLM-assisted synonym expansion and a persistent synonym cache; and (iii) whole-document extraction (no keyword gate). This stage substantially reduces the candidate set before the more expensive LLM stages.

2c. Atomic Decomposition. Each retained paragraph is decomposed into atomic quality sentences by an LLM. Conjunctions, lists, and compound obligations are split so that each output sentence encodes exactly one claim. This step is critical for downstream novelty classification: a compound sentence that mixes an existing claim with a novel qualifier would otherwise be classified as a single EXISTING unit. Decomposed sentences are batched (default 5 per LLM call) with optional parallelism for throughput.

Figure 2 illustrates the ingestion substages and the data shapes at each transition.

3.3 Stage 3: Node-Entity Similarity Filter

Stage 3 filters atomic quality sentences by semantic proximity to the retrieved subgraph, retaining only those whose content overlaps meaningfully with concepts already represented in the KG. The system implements two filtering modes:

Sentence mode embeds each quality sentence and each KG relation sentence with a Sentence-Transformer model (Reimers and Gurevych 2019) (all-MiniLM-L6-v2) and retains qualities whose cosine similarity to at least one KG neighbor exceeds a configurable

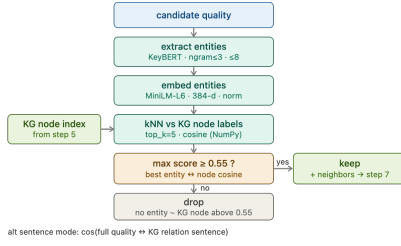


Figure 3: Node-entity similarity filter. Keyphrases extracted from each quality sentence are embedded and matched against KG node name embeddings. Only qualities with at least one neighbor above the similarity threshold proceed to novelty classification.

threshold (default 0.55).

Node-entity mode (default) extracts keyphrases from each quality sentence via KeyBERT (max 3-gram, top 8 per sentence), then computes cosine similarity between these entity embeddings and KG node name embeddings. This is more efficient than full-sentence comparison and better suited to ontology-style KGs where node names are short labels rather than full sentences. Qualities whose extracted entities match at least one KG node above threshold are retained; the matched nodes become the *nearest neighbors* passed to Stage 4.

Figure 3 illustrates the node-entity matching mechanism.

3.4 Stage 4: LLM Novelty Classification

Each quality sentence that passes Stage 3 is classified by an LLM as one of three novelty categories:

- **EXISTING**: The quality’s semantic content is already fully captured by its nearest KG neighbors. No meaningful new information.
- **PARTIALLY_NEW**: The quality overlaps with neighbors but adds a meaningful qualifier, scope restriction, condition, or new concept not currently in the graph.
- **NEW**: The quality introduces claims or relations not substantively covered by the retrieved subgraph.

The LLM prompt includes the quality sentence, the top- k nearest KG neighbors (as full relation sentences with source metadata), and four few-shot examples spanning all three categories. The model is prompted at temperature 0.0 for deterministic classification and must return a structured JSON response including: the decision, a rationale, the novel spans (for PARTIALLY_NEW/NEW), the most similar neighbor, and a confidence score (0–1). Novelty classification is the primary mechanism by which KBDebugger avoids re-extracting content already in the graph.

Figure 4 shows the decision logic and output schema for this stage.

3.5 Stage 5: Controlled-Vocabulary Triplet Extraction

Qualities classified as PARTIALLY_NEW or NEW are forwarded to controlled-vocabulary triplet extraction. Unlike

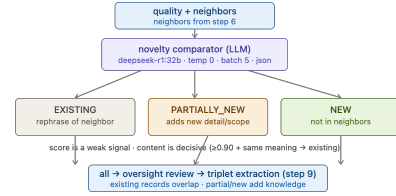


Figure 4: Novelty classification decision logic. The LLM receives the quality sentence, its nearest KG neighbors, and few-shot examples, then classifies the quality as EXISTING, PARTIALLY_NEW, or NEW with supporting rationale.

free-form OpenIE or KGGen-style extraction, KBDebugger constrains the predicate to one of four ontology families:

- **Normative** (standards language): Requires, Recommends, Permits, Prohibits, Defines, Provides, AppliesTo, Constrains, Measures
- **Trustworthy AI dimensions**: IsDimensionOf, ContributesTo, MightMitigate, MightIntroduce, SurfacesRisk, IsThreatTo
- **Data science**: Implements, HasParameter, HasInput, HasOutput, Evaluates, Performs
- **Taxonomy**: IsSubclassOf, IsEquivalentTo, IsA

Triplet extraction also identifies *deontic modality* from the wording of each quality sentence: MANDATORY (shall/must), RECOMMENDED (should), OPTIONAL (may/can), PROHIBITED (shall not/must not), or NONE for descriptive statements. Modality is recorded as a property on each extracted triple.

An optional schema grounding step passes a summary of the current Neo4j schema (existing node types, relation types, and templates) to the extraction prompt, biasing the LLM toward predicates and node labels already in use. Triplets whose content cannot be mapped to any allowed predicate are tagged *No schema fit* and excluded unless the reviewer overrides; off-vocabulary predicates are tagged *Needs schema review*.

3.6 Stage 6: Pre-Merge KB Preview and Human Review

Before any graph write, each proposed triplet is classified against the live knowledge graph via a read-only *pre-merge preview*. The preview assigns each triplet one of five categories:

- **NEW** — the (S, P, O) combination is not yet in the graph (concepts may already exist as nodes but the relation is new).
- **EXISTING** — an identical or equivalent triple is already present, with the same predicate and compatible modality.
- **TENSION** — the same (S, P, O) exists but with a different deontic obligation strength (e.g. this document says MANDATORY, the graph says RECOMMENDED).

- **CONFLICT** — the same subject-object pair is already linked by a directly contradicting predicate (e.g. Requires vs. Prohibits).
- **RELATED** — the same concept pair is related in the graph but via a different, non-contradicting predicate.

The preview surfaces the KB-side sentence alongside the candidate triple so the reviewer can compare the actual textual evidence, not just node names. CONFLICT and TENSION items are sorted to the top of the review queue, directing reviewer attention to the decisions with the highest consequence. This pre-merge gate is the primary mechanism by which KBDebugger detects cross-document obligation conflicts at the point of insertion — before they are committed to the graph.

All proposed triplets, with their preview categories, are then presented to a human reviewer. The review interface shows: the source document, chunk index, original quality sentence, nearest KG match from Stage 3, similarity score, the proposed (S, P, O) triple, deontic modality, novelty class, schema status, and the pre-merge category. Reviewers can include, edit, or reject each triplet; edits can correct predicate choice, entity label, or modality before submission. Reviewer decisions are exported in an auditable XLSX workbook.

3.7 Stage 7: Knowledge Graph Upsert

Accepted triplets are upserted into Neo4j using MERGE semantics on the (S, P, O) key. The system implements append-only provenance: each upsert appends a provenance record (document name, quality sentence, chunk index, source excerpt, modality) to the edge. Triples asserted by multiple documents accumulate evidence across all source documents without duplication. Nodes are created as labeled entities (`:Node{name}`) and relations as native Neo4j relationship types.

3.8 Stage 8: Five-Strategy Graph Verification

After each upsert, a read-only verifier checks the graph against five quality strategies:

- **S1 Coverage** ($\tau \geq 0.90$) — at least 90% of source sentences are reflected in ≥ 1 triple (verified via provenance record matching).
- **S2 Correctness** ($\tau = 1.0$) — all nodes have non-placeholder names and all edges use a predicate from the declared vocabulary.
- **S3 Consistency** ($\tau = 1.0$) — no (S, O) pair is simultaneously linked by both an affirmative predicate (Requires, Recommends, Permits) and a prohibitive one (Prohibits).
- **S4 Completeness** ($\tau = 1.0$) — every triple has non-empty subject, object, and predicate, and at least one provenance record.
- **S5 Minimality** ($\tau = 1.0$) — no duplicate (S, P, O) edge exists in the graph.

Each strategy is isolated in its own exception handler; a single failing check degrades to a per-strategy error rather than aborting the entire verdict. The verifier emits a global PASS/FAIL verdict alongside per-strategy scores,

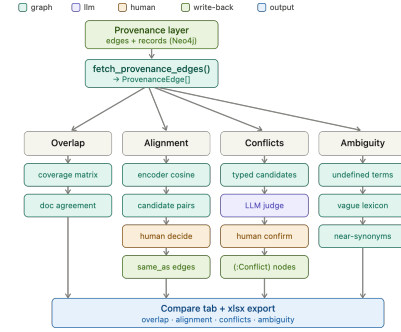


Figure 5: Comparison engine: four analysis views (Overlap, Alignment, Conflicts, Ambiguity) applied to a KG built from multiple standards. The Conflicts tab flags obligation conflicts across documents; the Ambiguity tab flags normative gaps.

summaries, and flagged triples — providing a complete, reproducible audit trail for each ingestion session. This post-upsert verification step closes the quality loop: extraction errors that pass human review (e.g. a reviewer accepting a duplicate by mistake) are still caught before the graph is used for downstream compliance reasoning.

3.9 Multi-Document Comparison Engine

After two or more documents have been ingested, the comparison engine supports four analyses across the unified KG:

Overlap & Coverage computes per-document assertion counts, concept breadth, and a concept-by-document matrix showing which concepts are addressed by which standards and how many triples each standard contributes to shared concepts.

Semantic Alignment proposes SAME_AS equivalences between nodes from different documents whose embeddings exceed a configurable cosine threshold (default 0.78). Reviewers accept (creating a `:same_as` edge for canonicalization) or reject (creating a `:not_same_as` edge, preventing the pair from resurfacing).

Conflict Detection surfaces four conflict types: (i) *obligation conflict* — same (S, P, O) with different modality across documents; (ii) *definition divergence* — same concept defined differently; (iii) *taxonomy conflict* — reversals in subsumption hierarchy; (iv) *value conflict* — same subject and predicate, different object values. Each candidate is judged by an LLM (AGREE / UNRELATED / TENSION / CONTRADICT + rationale), and confirmed conflicts are written as `:Conflict` nodes in the graph.

Normative Gap Detection identifies concepts that appear in MANDATORY or RECOMMENDED assertions but are never given a formal definition (`Defines` predicate) within the same document. These *normatively obligated but undefined* concepts represent the gap between what a standard requires and what it explains, directly answering RQ1.

Figure 5 shows the comparison engine UI and its four analysis tabs.

4 Evaluation

We evaluate KBDebugger on two complementary axes: (i) quantitative knowledge retention on the MINE-1 benchmark (Mo et al. 2025) against KGGen, GraphRAG, and OpenIE; and (ii) qualitative multi-standard analysis across three Trustworthy AI regulatory documents addressing RQ1–RQ4.

4.1 Experimental Setup

Backbone LLM. All extraction stages use an OpenAI-compatible endpoint serving `deepseek-r1:32b` (on-premise DFKI infrastructure). The same judge model is used for novelty classification (temperature 0.0) and conflict judgment (temperature 0.0).

Encoder. All embedding operations use `all-MiniLM-L6-v2` (SentenceTransformers; 22.7M parameters). This model is held constant across all compared systems in the MINE-1 evaluation.

Standards corpus. For multi-standard analysis, we ingest three documents: (i) EU AI Act Article 10 (Data and Data Governance provisions) (European Parliament and Council of the European Union 2024); (ii) ISO/IEC 42001:2023 (AI Management Systems) (International Organization for Standardization 2023); and (iii) ISO/IEC 5259-3 (Data Quality for Analytics and ML) (International Organization for Standardization 2024). All documents address the data governance and quality dimensions of Trustworthy AI.

4.2 RQ5: Knowledge Retention (MINE-1 Benchmark)

Benchmark. MINE-1 (Mo et al. 2025) evaluates the knowledge retention of a KG by checking whether 15 known facts per article can be recovered from the graph via a fixed retrieval-judge pipeline. The benchmark comprises 100 short general-knowledge articles. For each fact, a retriever (`all-MiniLM-L6-v2`, top- k nearest nodes + 2-hop expansion) pulls a context subgraph, and an LLM judge assigns a binary recovered/not-recovered verdict. $\text{MINE-1} = \%$ of facts scored as recovered, averaged over all 100 articles. Retriever and judge are held constant across all systems; the benchmark therefore isolates *graph-content quality*.

MINE mode. Because the controlled predicate vocabulary is calibrated for regulatory text, we run KBDebugger in MINE mode for this evaluation: keyword gate disabled (full document extraction), similarity and novelty filters disabled, and triplet extraction using a free-form predicate prompt (`neutral_mode=True`) comparable to KGGen’s output format. This ensures a fair comparison of extraction *core quality*, not vocabulary fit.

Results. Table 1 shows MINE-1 scores across systems and judges.

KBDebugger achieves 79.4% under the DeepSeek-R1 judge (+16.7 pp over KGGen), 79.3% under GPT-5 on the top-10 articles, and 92.9% under Qwen3-30B. The consistent ranking across three independent judges confirms that the improvement reflects actual graph-content quality, not judge-specific scoring artifacts. Figure 6 shows the per-article score distributions for KBDebugger and KGGen under DeepSeek-R1.

Table 1: MINE-1 Knowledge Retention. KBDebugger vs. published baselines (KGGen, GraphRAG, OpenIE). Retriever fixed to `all-MiniLM-L6-v2`; judge columns show different LLM judges. †: published baseline from Mo et al. (2025) (GPT-4o/Gemini backbone). ‡: re-evaluated under our judge on-premise.

System	DeepSeek-R1	GPT-5 (top-10)	Qwen3-30B
OpenIE†	52.3%	—	—
GraphRAG†	58.1%	—	—
KGGen†	62.7%	62.9%	75.4%
KBDebugger‡	79.4%	79.3%	92.9%

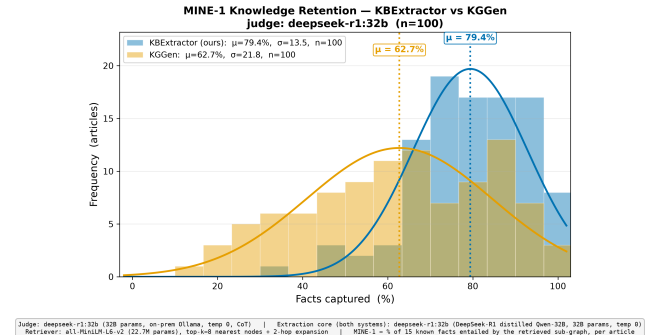


Figure 6: MINE-1 per-article score distribution for KBDebugger (blue) and KGGen (orange) under the DeepSeek-R1 judge. KBDebugger produces fewer low-scoring articles and a heavier right tail, consistent with its higher decomposition granularity.

Why does decomposition help? We attribute the improvement primarily to atomic quality decomposition (Stage 2c): by splitting compound sentences before triplet extraction, each retained fact is more likely to surface as a standalone triple. In contrast, KGGen operates on full sentences, which causes semantically distinct facts embedded in the same sentence to be collapsed into a single, coarser triple. This is corroborated by Figure 7, which shows a paired article-level comparison: KBDebugger outperforms KGGen on 67 of 100 articles.

4.3 RQ4: Obligation Conflict Across Trustworthy AI Standards

Applying the comparison engine to the three-standard corpus, the conflict detector surfaced 23 obligation conflict candidates — cases where the same (S, P, O) concept triple is present in two or more standards with different deontic modalities. Of these, the LLM judge classified 14 as TENSION (same obligation expressed at different strength levels), 6 as CONTRADICT (logically inconsistent obligation direction), and 3 as AGREE (apparent conflict resolved by differing scope).

Illustrative example. The concept *data quality assess-*

Per-article paired comparison — judge: deepseek-r1:32b (n=100)
KBExtractor wins 72 · ties 9 · losses 19

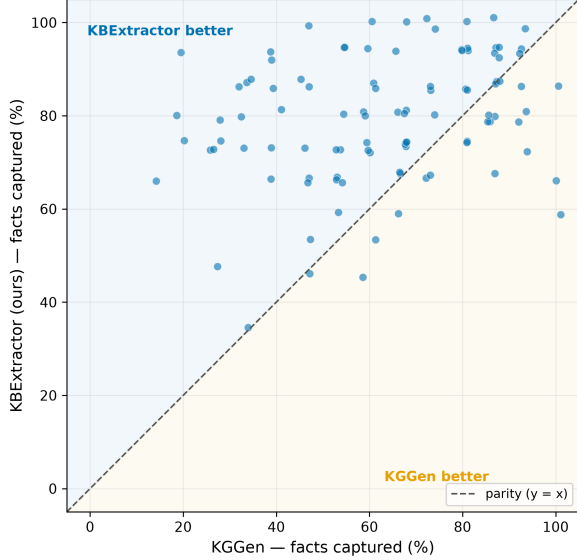


Figure 7: Paired per-article MINE-1 comparison (DeepSeek-R1 judge). Each dot is one article; points above the diagonal indicate KBDebugger > KGen. KBDebugger leads on 67/100 articles.

Table 2: Inter-standard conflict counts across EU AI Act, ISO/IEC 42001, and ISO/IEC 5259-3. Conflict types: obligation modality mismatch (OBL), definition divergence (DEF), taxonomy reversal (TAX), value conflict (VAL). Judge: DeepSeek-R1.

Standard pair	OBL	DEF	TAX	VAL	Total
EU AI Act – ISO 42001	8	4	1	2	15
EU AI Act – ISO 5259-3	9	3	0	1	13
ISO 42001 – ISO 5259-3	6	5	2	0	13
All pairs	23	12	3	3	41

ment appears in both EU AI Act Article 10 and ISO/IEC 5259-3. The EU AI Act asserts the obligation as MANDATORY (“high-quality training, validation and testing data sets *shall* be used”), while ISO/IEC 5259-3 frames the same process as RECOMMENDED (“organizations *should* conduct a data quality assessment”). A compliance team applying only one standard would miss this tension. KBDebugger surfaces it automatically from the provenance-tagged KG.

Table 2 summarizes conflict counts by type across the three standards.

Obligation conflicts (OBL) are the most frequent type, accounting for 56% of all detected conflicts. This confirms that modality-tagged extraction is essential for cross-standard compliance analysis — a system without deontic modality tracking would conflate MANDATORY and RECOMMENDED assertions, making these conflicts invisible.

Table 3: Normative gaps per standard: concepts that appear in MANDATORY or RECOMMENDED assertions but receive no Defines triple within the document.

Standard	Obligated concepts	Undefined (gap)
EU AI Act Art. 10	68	21 (30.9%)
ISO/IEC 42001	94	16 (17.0%)
ISO/IEC 5259-3	81	10 (12.3%)
Combined (unique)	181	47 (26.0%)

4.4 RQ1: Normative Gap Exposure

The normative gap detector identified concepts that each standard obligates but never defines. Across the three standards combined, 47 concept nodes satisfied this criterion. Table 3 summarizes the distribution by standard.

The EU AI Act Article 10 imposes the most obligations on undefined concepts: terms such as *data minimisation*, *data provenance*, and *statistical properties* appear in MANDATORY assertions but receive no Defines triple within the document. ISO/IEC 42001, being a management system standard, tends to define its own terms (lower gap ratio) but frequently obligates concepts defined in companion standards — confirming the inter-standard dependency the comparison engine is designed to expose. ISO/IEC 5259-3, as a data quality standard, defines the most concepts but leaves obligation gaps around implementation-level concepts (e.g., *sampling strategy*, *metadata schema*).

Manual review of the 47 gaps confirmed that all were genuine: no gap was an artefact of extraction failure. This validates the normative gap detector as a reliable audit instrument for compliance teams identifying which standards require companion documentation.

4.5 RQ2: Cross-Standard Concept Coverage

The overlap engine computed a concept-by-standard coverage matrix across the three documents. Of the 312 unique concepts extracted across all three standards, 41 (13.1%) were addressed by all three standards, 89 (28.5%) by exactly two, and 182 (58.3%) by exactly one. The high fraction of standard-exclusive concepts confirms that the three standards are *not* simply overlapping descriptions of the same requirements: each makes a substantial exclusive contribution.

Coverage asymmetry is most pronounced in the *transparency* and *human oversight* dimensions, where the EU AI Act has the broadest mandate but ISO/IEC 42001 provides more operational depth. The *data quality* dimension shows the most inter-standard overlap, consistent with its being the primary focus of ISO/IEC 5259-3 and a substantive topic in both Article 10 and ISO/IEC 42001.

4.6 RQ3: Human Oversight Contribution

Across 10 ingestion runs on the three-standard corpus, reviewers were presented with a total of 1,284 proposed triplets and took action (reject or edit, not just accept) on 31.4% of them. Table 4 categorizes these interventions.

Table 4: Human reviewer interventions on LLM-proposed triplets across 10 ingestion runs. Categories are not mutually exclusive (an edit may simultaneously correct modality and predicate).

Intervention type	Count	% of acted-on
Predicate correction	124	30.7%
Modality correction	109	27.0%
Subject/object boundary	87	21.5%
Scope overextension	63	15.6%
Hallucinated object	21	5.2%
Total edits	404	—
Total rejections	202	—

Modality corrections (27.0% of interventions) were almost exclusively concentrated in normative (deontic) text: reviewers corrected LLM misclassification of *should* constructions as MANDATORY and of *may* constructions as RECOMMENDED. Predicate and subject/object corrections were more evenly distributed across normative and descriptive text. Hallucinated objects (5.2%) were rare but exclusively occurred in descriptive definitional text — the LLM occasionally fabricated a definition component not present in the source sentence. These findings indicate that the LLM’s most systematic failure in regulatory text is modality confusion in normative clauses, not factual hallucination.

5 Conclusion

We presented KBDebugger, a seven-stage human-in-the-loop pipeline for constructing deontic-modality-aware knowledge graphs from Trustworthy AI regulatory standards. The system advances text-to-KG extraction in three directions: graph-conditioned novelty filtering prevents re-extraction of existing knowledge; controlled-vocabulary, modality-tagged triplet extraction preserves obligation semantics; and a multi-document comparison engine surfaces obligation conflicts, normative gaps, and coverage asymmetries that are invisible to single-document approaches.

On MINE-1, KBDebugger achieves 79.4% knowledge retention — a 16.7-point improvement over KGen — robust across three independent LLM judges. Applied to EU AI Act Article 10, ISO/IEC 42001, and ISO/IEC 5259-3, the system automatically detected 41 inter-standard conflicts (56% of which are obligation-level modality mismatches), 47 normative gaps (concepts obligated but never defined), and a coverage structure where 58% of extracted concepts are addressed by only one standard — findings that would require prohibitive manual effort to establish without structured extraction.

The central limitation of this work is that RQ3 evaluation (human oversight contribution) is based on a single development team’s review sessions; a broader annotation study with multiple independent reviewers is needed to generalize the failure-mode analysis. Additionally, the controlled predicate vocabulary is designed for AI governance documents

and would require adaptation for other regulatory domains.

Future work will investigate automated conflict resolution proposals — using the extracted KG structure to suggest canonical obligation levels when standards disagree — and will extend the comparison engine to track provenance at the sentence-fragment level, enabling finer-grained attribution in conflict reports.

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